Main results

Positive Summation Kernels

A family of functions $(K_n)_{n\geq 1}$ on $\mathbb{R}$ (or on $\mathbb{T}$, the circle) is called a *positive summation kernel* if the following hold:

(i) $K_n(x) \geq 0$ for all $x \in \mathbb{R}$ and each n;

(ii) $\int_{-\infty}^{\infty} K_n(x)\,dx = 1$ for all n;

(iii) for every $\delta > 0$,

$$\int_{|x|\geq\delta} K_n(x)\,dx \longrightarrow 0 \quad \text{as } n \to \infty.$$

Equivalently, (K_n) is an *approximate identity*: the mass of K_n concentrates near $x = 0$ as $n \to \infty$. For such kernels, the convolution $(f * K_n)(x) = \int f(x-t)K_n(t)\,dt$ satisfies

$$(f * K_n)(x) \to f(x)$$

for all x where f is continuous.

Riemann-Lebesgue Lemma

Theorem 2.2 (Riemann–Lebesgue lemma) Let f be absolutely integrable in the Riemann sense on a finite or infinite interval I. Then

$$\lim_{\lambda\to\infty} \int_I f(u)\sin(\lambda u)\,du = 0.$$

(A corresponding result holds for $\cos(\lambda u)$ or $e^{i\lambda u}$.)

Corollary for Fourier Coefficients

Corollary. Let $f \in L^1(\mathbb{T})$ (i.e., $\int_{-\pi}^{\pi} |f(x)|dx < \infty$). Its complex Fourier coefficients are:

$$\hat{f}(n) = c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}\,dx, \qquad n \in \mathbb{Z}.$$

The Fourier coefficients of any integrable function vanish at infinity:

$$\hat{f}(n) \longrightarrow 0 \quad \text{as } |n| \to \infty.$$

Proof Sketch of Corollary: The Fourier coefficient $\hat{f}(n)$ can be expressed as:

$$\hat{f}(n) = \frac{1}{2\pi}\left(\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx + i\int_{-\pi}^{\pi} f(x)\sin(-nx)\,dx\right)$$

Since $\cos(nx)$ and $\sin(nx)$ are bounded functions, the absolute integrability of f is preserved when multiplied by these functions. As $|n| \to \infty$, both the real and imaginary parts of the integral tend to zero by applying the Riemann–Lebesgue lemma (and its cosine variant) with $\lambda = n$. Therefore, $\hat{f}(n) \to 0$ as $|n| \to \infty$.

Fourier Series Coefficients

The representation of a 2π-periodic function f as a Fourier series depends on whether the complex exponential basis or the real trigonometric basis is used.

1. Complex Fourier Series

For a function $f \in L^1(\mathbb{T})$, the series representation is:

$$f(x) \sim \sum_{n\in\mathbb{Z}} c_n e^{inx}$$

The complex Fourier coefficients $c_n = \hat{f}(n)$ are computed as:

$$c_n = \hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}\,dx, \qquad n \in \mathbb{Z}.$$

2. Real Fourier Series (Trigonometric)

For a real-valued function f, the series representation is:

$$f(x) \sim a_0 + \sum_{n=1}^{\infty}(a_n\cos(nx) + b_n\sin(nx))$$

The real Fourier coefficients are computed as:

- a_0 (Constant component):
$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$
- a_n (Cosine coefficients, $n \geq 1$):
$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \qquad n \in \mathbb{N}$$
- b_n (Sine coefficients, $n \geq 1$):
$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx, \qquad n \in \mathbb{N}$$

Note: The relationship between the coefficients is $c_0 = a_0$, and for $n \geq 1$, $c_n = \frac{1}{2}(a_n - ib_n)$ and $c_{-n} = \frac{1}{2}(a_n + ib_n)$.

Plancherel's Theorem (Parseval's Identity for Fourier Transforms)

Plancherel's Theorem relates the L^2 norm of a function in the time domain $L^2(\mathbb{R})$ to the L^2 norm of its Fourier Transform in the frequency domain $L^2(\mathbb{R})$. This identity is essential for evaluating integrals using Fourier transforms.

1. Squared L^2-Norm

For a square-integrable function $f \in L^2(\mathbb{R})$, the L^2-norm is preserved under the Fourier Transform:

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi$$

Note: The factor $\frac{1}{2\pi}$ depends on the convention used for the Fourier Transform.

2. Polarized Plancherel Formula

For $f, g \in L^2(\mathbb{R})$, the inner product is preserved:

$$\int_{-\infty}^{\infty} f(t)\overline{g(t)} dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\xi)\overline{\hat{g}(\xi)} d\xi$$

The integrals in Problem 6(b) often rely on applying these formulas, usually by identifying the integrand with either the square of a transform $|\hat{f}(\xi)|^2$ or a product of transforms $\hat{f}(\xi)\overline{\hat{g}(\xi)}$.